

Business System Analyst

Prithvi Pendem

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PROFESSIONAL SUMMARY

- 9+ years of Experience as a **Senior Business System Analyst** in **Health Care domain** dedicated to continuous learning and staying abreast of emerging technologies to enhance analytical capabilities and provide innovative solutions.
- Having extensive domain knowledge on healthcare data privacy and security regulations, with a comprehensive understanding of **HIPAA, PPACA, HITECH** requirements and their impact on data management practices.
- Knowledge on claims process and adjudication in the Medicare Part A, B, C and D, Medicaid, Managed care, Private and TPA, PPO, HMO, and POS.
- Conducted end-to-end testing, including functional, performance, security, and usability tests for web applications, ensuring compliance with regulations such as HIPAA, HITECH, PPACA.
- Experience in Healthcare **EDI transaction** sets like 837, 835, 270/271, 276/277 & 278 and experience in translation of EDI transactions (837,834) to **Facets** Gateway format and also ability to understand carrier EDI or data companion guidelines, leveraging standards like **ANSI X12, EDIFACT**
- Extensive expertise in **FHIR** standards, resources, profiles for seamless data exchange and interoperability between **EHRs, EMRs, HIEs**, and healthcare systems, utilizing tools like **Rhapsody** and **Mirth Connect** for FHIR-based integrations and data transformations.
- Strong Background in **ICD-10/CPT/HCPCS/LOINC/SMONED CT** coding by maintaining knowledge of appropriate coding procedures to ensure correct coding of medical records and processing claims.
- Spearheaded a comprehensive **LIMS** integration, ensuring **CLIA** compliance, leveraging **LOINC** and industry coding standards for seamless data interoperability with **EHRs**.
- Expertise in understanding and supporting the client with Project Planning, Project Definition, Requirements Definition, Analysis, Design, Testing, System documentation and user training.
- Excellent working Knowledge of various Software Development Life Cycle (**SDLC**) methodologies like **Waterfall Model, Agile (SCRUM, Kanban)**, utilizing **JIRA** for project management and **Confluence** for collaborative documentation throughout the SDLC process.
- Strong experience in Data Analysis, Migration, Cleansing, Transformation, Integration, Data Import, and Data Export through the use of multiple **ETL** tools such as **Talend** and **Informatica PowerCenter**.
- Expertise in **Source-to-target mappings**, and meticulous documentation of data flow between various systems, **databases** and **data warehouses**.
- Excellent in designing business models using **UML** diagrams such as use case diagrams, class diagrams, activity diagrams, sequence diagrams and collaboration diagrams using **Lucid Chart** and **MS Visio**.
- Proficient in Data Mapping, Modeling, Star/Snowflake Schemas, and E-R Model design. Experienced in **OLTP/OLAP** systems, ensuring efficient data organization and retrieval for informed decision-making.
- Experienced in conducting **Joint Application Development (JAD)** sessions, **white board sessions**, project meetings with cross-functional teams, including end-users, Subject Matter Experts (SMEs), developers, and QA and written Use cases, User stories, test plans and test cases.
- Conducted **Gap Analysis, Root Cause Analysis** and **Impact Analysis, SWOT Analysis** to identify discrepancies, underlying issues and assess the consequences, enabling informed decision-making.
- Acted as a liaison between Business and Technical Stakeholders to elicit various **Functional and Non-Functional** requirements documents using **Adobe XD**.
- Successful Leadership in **User Acceptance Testing (UAT)** efforts across multiple projects, including devising test plans, scenarios and test cases to validate system functionality using **Zephyr**.
- Proficient in the **Microsoft Office** Suite, encompassing Excel, Word, PowerPoint, and Outlook. Utilizes advanced features and functions to adeptly analyze data, craft comprehensive reports and presentations, and facilitate seamless communication with stakeholders.

- Skilled in developing comprehensive documentation, including business requirement documents (**BRDs**), functional requirement documents (**FRDs**), and user stories, ensuring clear communication between stakeholders and development teams.
- Expertise in creating **dashboards** and **stories** in **Tableau** and **QlikView**, utilizing advanced features like calculated fields, parameters, calculations, groups, sets, hierarchies, LOD expressions, and localization.
- Proficient in leveraging **Power BI**'s comprehensive suite of tools, including Power View, Power Query, Power Pivot, and Power Maps, to design and develop visually compelling **dashboards**, **scorecards**, and **KPI** reports that provide actionable insights and drive informed decision-making.

TECHNICAL SKILLS

- **Computer Skills:** Data visualization, Data Cleaning, Data Migration, MS Office (Access, Word, Excel, Power point), Cerner Applications such as Citrix workspace, Relay, Microsoft teams, problem solving and analytical abilities, Mathematical and statistical skills.
- **Analytical Tools:** Tableau Desktop, MS-Excel, Power BI, Qlik View.
- **Languages:** SQL, SAS, Python, R, XML.
- **Cloud services skills:** AWS, Google Cloud Platform (GCP), Microsoft Azure and Data Bricks.
- **Project Management Tools:** JIRA, Trello.
- **ETL Tools:** Informatica, Talend.
- **Data Modeling:** Star-schema and Snowflake schema, FACT and dimensions, Pivot Table.
- **Databases:** Oracle, MS Access, SQL.
- **Operating systems:** Windows, LINUX, Mac OS and DB2.
- **Expertise:** Electronic Health Records, Electronic Medical Records, EDI Transactions, HL7, FHIR, LIMS, EDC, ELN and other healthcare operations
- **Healthcare Analytics:** Knowledge of healthcare tools and HIPAA compliance.
- **Requirements Gathering:** Documenting and prioritizing business requirements.

PROFESSIONAL EXPERIENCE:

Senior Business System Analyst
Reckitt, Evansville, IN

Mar 2022 - Present

- **Responsibilities:**
- Collaborated with cross-functional teams to analyze and optimize Electronic Data Interchange (EDI) transactions, including (837, 835, 270, 271, 276, 277, and 278), resulting in streamlined claims processing workflows, enhanced efficiency, and improved accuracy.
- Developed comprehensive documentation, including functional specifications, technical workflows, and user manuals, to support the implementation of complex software systems.
- Led requirements gathering sessions with stakeholders, identifying key technical and business needs to inform software development projects.
- Developed detailed source-to-target mapping (STTM) documents for ETL processes across data lakes and warehouses.
- Developed HiveQL scripts to extract and transform large volumes of unstructured data for downstream analytics.
- Implemented complex joins, aggregations, and window functions using Hive for customer behavior analysis.
- Designed and implemented validation logic for 837 institutional (837I) and professional (837P) claims to reduce errors in Medicaid and Medicare Advantage plan submissions.
- Collaborated with eligibility vendors to enhance EDI 270/271 real-time eligibility responses, improving member access to services across Medicaid and CHIP populations.
- Refined 835 remittance advice EDI process to ensure accurate posting of claim payments and denials for commercial and exchange-based health plans.

- Participated in cross-functional Agile teams to prioritize and implement EDI backlog items related to CMS compliance and encounter data submissions.
- Coordinated with enrolment teams to ensure consistency between EDI eligibility (834) files and claims processing systems for ACA and Medicaid beneficiaries.
- Implemented improvements in rejected claim tracking for retail pharmacy transactions, integrating dashboards to monitor denial reasons and resolution paths.
- Mapped source-to-target data elements for pharmacy claim EDI files, including diagnosis codes, NDC, DUR responses, and prescriber details.
- Led initiatives to ensure EDI compliance for Medicare Part D claims under CMS guidelines, improving audit readiness and reducing compliance risks.
- Conducted analysis and cleanup of legacy pharmacy EDI files during migration to new PBM or payer systems, ensuring continuity in claims processing.
- Coordinated pharmacy EDI enhancements related to value-based contracting and specialty pharmacy performance monitoring.
- Ensured compliance with regulatory standards such as the **HIPAA**, **HITECH** Act, and **CMS** guidelines.
- Collaborated with cross-functional teams to streamline the EDI transaction flow for Medicare, Medicaid, and Managed Care claims processing. Analyzed and optimized the processing of **CMS-1500** and **UB-04** forms.
- Contributed to enhancing software **UI/UX** by collaborating with developers, utilizing Sketch and Figma to design intuitive and visually appealing interfaces, improving user satisfaction and interaction efficiency.
- Designed and developed interactive wireframes and mockups using Balsamiq to visualize user interfaces for healthcare and enterprise applications.
- Proficient in handling API responses, error management, and ensuring API scalability and performance. Experience in versioning and managing APIs to ensure compatibility across various system versions.
- Documented project processes and procedures using **Rational ClearCase** and created **artifacts** and milestones for various phases of the SDLC.
- Developed and deployed scalable data pipelines on **Databricks** using Apache Spark, optimizing EDI transaction processing for claims submissions and adjudications.
- Embraced **Agile** methodologies, including **Scrum framework** and **JIRA** tool, to prioritize **tasks** in **Product backlog**, foster collaboration, and streamline project management.
- Created and managed User Stories, Epics, and Tasks in **JIRA** to track project progress and ensure alignment with business objectives and requirements.
- Utilized ADO dashboards, queries, and work item tracking to monitor project progress, sprint backlogs, and system enhancements for the On-Exchange Individual marketplace.
- Created and maintained workflows, backlog items, and sprint planning boards in Azure DevOps, ensuring efficient tracking of business and technical requirements.
- Defined and prioritized product backlogs, ensuring alignment with business objectives and customer needs.
- Facilitated sprint planning sessions, ensuring clear communication of priorities and setting achievable goals for each iteration.
- Led backlog grooming sessions, collaborating with stakeholders and development teams to refine and prioritize the product backlog.
- Developed wireframes and interactive UI mockups to communicate design concepts and functionality to stakeholders and development teams.
- Created and maintained Salesforce reports and dashboards for actionable business insights and decision-making.
- Facilitated Joint Application Requirements (**JAR**) and Joint Application Development (**JAD**) sessions with stakeholders to gather and document business and system requirements, ensuring alignment with project goals and user needs.
- Participated in group sessions to define and validate data migration requirements, identify process gaps, and ensure alignment with system needs.
- Developed **Informatica PowerCenter** ETL processes to integrate and transform healthcare data from EDI transactions, claims databases, medical records, and other sources into the data warehouse.

- Developed Use Cases, Activity Diagrams, Sequence Diagrams, and Object-Oriented Designs (OOD) using **UML** and **Visio**, worked closely with development teams to incorporate **use cases** into the design and implementation phases.
- Maintained a **Traceability Matrix** to systematically link functional requirements, use cases, and test cases, ensuring comprehensive coverage and effective requirements management throughout the software development lifecycle.
- Implemented comprehensive user acceptance testing (**UAT**) protocols to validate the effectiveness and accuracy of optimized EDI transaction processes and software solutions.
- Performed back-end **data validation** testing using **SQL queries** against Oracle, Access and Postgres databases and created **stored procedures** as needed for data processing, and identified data anomalies using SQL queries.
- Tested performance and accuracy related **queries** using SQL Server, ensuring data integrity and system reliability. Implemented stored procedures and indexes in **SQL Server** to streamline data retrieval and manipulation processes, improving overall system efficiency.
- Utilized SQL Server Reporting Services (**SSRS**) to generate insightful reports on EDI transaction performance, enabling stakeholders to make data-driven decisions for process optimization.
- Developed complex **SQL queries**, **stored procedures**, and **triggers** to efficiently retrieve, manipulate, and update data in relational databases.
- Conducted **Gap analysis** of claims and enrollment processes to streamline HIPAA EDI X12 transactions, ensuring compliance and optimizing efficiency in healthcare claims processing.
- Developed interactive **Power BI** dashboards and reports to provide actionable insights and facilitate data-driven decision-making.
- Utilized **Power Query** to extract, transform, and load (ETL) data from various sources, ensuring data integrity and accuracy.
- Proficient in **Microsoft Office** Suite, including Word, Excel, and PowerPoint, for creating and delivering comprehensive reports, analyses, and presentations to support data-driven decision-making and effective communication across teams.
- Leveraged **MS-Excel** functions such as **VLOOKUP**, **pivot tables**, and other advanced Excel functions for data analysis.

- **Senior Business System Analyst**

May 2021- Feb 2022

- **State of Illinois, Illinois**

- **Responsibilities:**

- Spearheaded the end-to-end modernization of legacy **Claims EDI processing systems**, focusing on **837/835** formats and real-time integration. Led the extraction and analysis of claims lifecycle data to identify gaps in adjudication. Ensured compliance with **HIPAA** and **CMS** guidelines throughout all phases.
- Authored detailed **BRDs**, **FRDs**, and **data mapping documents** that aligned business expectations with technical deliverables. Facilitated workshops with claims processors, compliance officers, and data architects to capture system requirements. Maintained full traceability using **RTMs** across user stories and test cases.
- Developed and executed complex **SQL queries** for claims validation, membership matching, and duplicate detection. Automated data profiling routines to ensure data accuracy across all **ETL pipelines**. Provided validation checkpoints and summarized data issues for triage.
- Designed **Power BI dashboards** to visualize claim submission errors, denial trends, and reimbursement timelines. Delivered **role-based reporting views** for executives, finance, and compliance stakeholders. Enabled real-time data refreshes using **Power BI Gateway** and scheduled refresh policies.
- Collaborated with data engineers to ingest structured and unstructured claims data into **Databricks**. Applied **Spark SQL** transformations to normalize and clean the data prior to reporting. Validated transformations against source systems to ensure consistency.
- Used **HiveQL** within the **Hadoop** environment to query large claims datasets for anomaly detection and historical trend analysis. Scheduled jobs using **Control-M** for daily batch processing and dashboard population. Monitored job performance and implemented failure alerting mechanisms.
- Managed data integration across **Azure Data Lake**, **Azure SQL Database**, and internal data warehouse systems.

- Defined **data lineage** diagrams using **Visio** to track the movement of sensitive claims and enrollment data. Ensured alignment with internal security and governance policies.
- Automated test scenarios and data reconciliation rules for **System Integration Testing (SIT)** and **User Acceptance Testing (UAT)**. Collaborated with QA teams to write and validate over 150 test cases. Logged defects and coordinated resolution via **Azure DevOps**.
- Facilitated **Agile ceremonies** including sprint planning, backlog grooming, and daily standups. Maintained velocity tracking and ensured cross-functional alignment across developers, QA, and business leads. Used **Jira** and **Confluence** to manage stories, documentation, and retrospectives.
- Supported version control, code review, and build automation using **Git** and **Jenkins**. Coordinated deployment readiness with DevOps engineers and release managers. Ensured clean promotion of code to staging and production environments.
- Defined and implemented **data governance** controls including field-level encryption, **metadata management**, and business glossary curation. Collaborated with data stewards to standardize **KPI definitions** and reporting thresholds. Conducted periodic audits to validate adherence.
- Partnered with compliance and fraud teams to identify high-risk transactions and implement **AML** detection logic. Modeled suspicious patterns based on transaction thresholds, member activity, and provider relationships. Provided insights via Power BI reports with embedded drilldowns.
- Created and maintained **process flow diagrams** to document current-state and future-state workflows of claims ingestion. Illustrated key data touchpoints, decision logic, and manual interventions. Shared visuals during stakeholder alignment sessions and documentation reviews.
- Led the **training and onboarding** of business users and junior analysts through interactive workshops. Developed user manuals, SOPs, and quick start guides for Power BI, Azure DevOps, and data extraction processes. Collected feedback to improve training effectiveness and adoption.
- Achieved a 30% reduction in claim processing time and improved claims data quality by over 40%. Delivered the project **2 weeks ahead of schedule**, receiving commendation from senior leadership. Enabled full traceability and compliance reporting with minimal manual intervention.

Business System Analyst
Infor, Hyderabad, India
Responsibilities:

Feb 2017 - May 2019

- Provided technical support for commerce tools in the quote-to-order process, resolving issues and improving system stability.
- Developed automated reporting and monitoring systems dashboard to track and minimize the impact of priority issues.
- Created context diagram and data flow diagrams, analyzed cybersecurity data, automated reports, scheduled jobs sourced data into excel from python.
- Performed data merging, cleaning, and quality control procedures by programming data object rules into a database management system.
- Worked on Hadoop-based data solutions for large-scale analytics, improving reporting efficiency.
- Delivered strategic business rules, Install Base forecasting by predictive goals, writing complex **SQL** queries to fetch and process data.
- Designed and implemented testing strategies to improve data accuracy and system performance for commerce tools.
- Migrate data from traditional database systems to Azure databases.
- Create automated solutions using Data bricks, Spark, Python, Snowflake, HTML.
- Created and managed Epics, features, user stories to track project lifecycle.
- Skillfully employed various workarounds to meet functional and non-functional requirements.
- Collaborated with business users to analyze requirements, determine scope of projects, and address concerns of business questions.

- Provided insights and supported the development and implementation of business initiatives, systems, and/or processes. Data Cleaning and data dictionaries and Statistical reports and hypothesis reports were generated.
- Developed and implemented Key Performance Indicators (KPIs) and dashboards to visualize business trends, enabling data-driven decision-making and strategic planning.
- Used **PI-SQL** and **SQL** to create, run custom queries and reports to generate data for reports.
- Created **UNIX** scripts and environment files to run batch jobs.
- Participated in sprint planning and execution, backlog management, team collaboration, process improvement, risk management.
- Facilitated daily stand-ups, utilizing Agile tools like Jira, and quantifiable results from Agile implementations.
- Created and optimized the data model, streamlined their data integrity and data cleaning process.
- Automated sales commission reports created slicers and visualization using the Power Pivots, VLOOKUP and trending summaries dashboard per the clients' requirements.
- Exporting data into BI tool from multiple databases and ERP system to develop analytics of products/sales/shipping.
- Maintained the Splunk software to automatically send out an alert to notify the appropriate authority through email and activate the necessary support.
- Developed data mapping documentation to establish relationships between source and target tables including transformation processes using **SQL**.

Junior Business System Analyst
Tricon Infotech, IN

June 2012 - Jan 2017

Responsibilities:

- Conducted a comprehensive analysis of existing systems, including **LIMS** and Electronic Health Records (EHRs), to identify data interoperability requirements and opportunities, focusing on **CLIA** compliance, **HIPAA** regulations, and industry coding standards such as **LOINC**, **CPT**, **ICD**. This analysis included performing a **SWOT** evaluation to inform strategic decision-making and risk mitigation strategies.
- Utilized Postman's testing features to automate API testing, developing test scripts that validated response data, status codes, and performance benchmarks to ensure API reliability and functionality.
- Collaborated with IT teams and subject matter experts to design and implement data integration strategies within the LIMS, following the **Agile** SDLC methodology, including requirements gathering, system design, development, testing, and deployment phases.
- Generated detailed **JIRA** reports and analytics to monitor project performance, identify bottlenecks, and recommend process improvements.
- Created custom **JIRA** filters and saved searches to streamline task management and enhance team productivity.
- Proficient in integrating SaaS applications with on-premises systems and third-party services via APIs to ensure seamless data exchange.
- Utilized **SQL queries** to extract, transform, and load (ETL) data within the LIMS, optimizing data quality and integrity for interoperability purposes.
- Utilized **SQL Server** to design and implement complex queries, stored procedures, and triggers to streamline data integration and automate routine processes within the LIMS, enhancing data retrieval efficiency and system performance.
- Leveraged **Python** scripting for custom data transformations and automation tasks, enhancing the functionality and efficiency of data integration processes.
- Developed **Tableau dashboards** and **reports** to visualize laboratory data, including charts and graphs to monitor key performance indicators and compliance metrics, enabling stakeholders to gain insights and make informed decisions.
- Developed **Tableau** datasets and dataflows to streamline data preparation and ensure consistency across reports and dashboards.
- Documented all aspects of the project, including specifications, data mappings, workflows, and technical documentation, following industry best practices and SDLC methodologies.

- Collaborated with IT teams and vendors to integrate **Electronic Lab Notebooks** (ELN) into the interoperability framework, streamlining laboratory workflows and documentation processes.

Education

GITAM University Electronics and Communication Engineering 2012

Texas Tech University Master of Engineering **2021**